



Approximating bandlimited signals associated with the LCT domain from nonuniform samples at unknown locations[☆]

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ARTICLE INFO

Article history:

Received 1 June 2011

Received in revised form

11 November 2011

Accepted 29 December 2011

Available online 8 January 2012

Keywords:

Nonuniform sample

Non-stationary signal

Linear canonical transform series

ABSTRACT

The sampling theory describes ways of reconstructing signals from their uniform or nonuniform samples associated with the traditional Fourier transform (FT). Most of the published papers about the sampling theory require signals to be bandlimited in the FT domain and assume that the sample locations and the band width are all known. However, the sample locations are not always known and most of the signals are non-stationary in practical applications. In order to overcome these shortcomings, this paper provides an algorithm for approximating signals from nonuniform samples at unknown locations. These signals are not necessarily bandlimited in the FT domain, however bandlimited in the LCT domain. The experimental results are given to verify the accuracy of the algorithm.

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1. Introduction

The sampling theory is one basic and important topic in engineering sciences, for it describes ways of reconstructing bandlimited signals from their samples. Uniformly sampled signals can be reconstructed via the Whittaker–Kotelnikov–Shannon sampling theorem [1]. For nonuniform samples, there are both iterative methods and non-iterative methods to recreate original signals [2–4]. But all of these methods always require signals to be bandlimited in the traditional Fourier transform (FT) domain and assume that the sample locations and the band width are all known. However, the sample locations are not always known, for instance, when the samples are collected with a jittered sample clock or when there is a certain unknown shift in a periodic nonuniform set of samples [5,6].

In [5], an effective algorithm was presented to recover unknown sample locations of a discrete-time bandlimited

signal. The method of approximating signals from nonuniform continuous-time samples at unknown locations was described in [6]. It is shown that these algorithms can work efficiently and have received much interests in signal processing community. However, those problems were all solved in the FT domain, i.e., the processed signal must be bandlimited in the FT domain. In practical applications, there are lots of signals which are non-bandlimited in the FT domain, however bandlimited in the linear canonical transform (LCT) domain [7]. For these signals, the above methods cannot be used, for it may lead to wrong (or at least suboptimal) conclusions. Under such circumstance, a natural idea is to try solving the problem in the LCT domain.

The LCT [8–10], which was introduced during 1970s with four parameters, has been proved to be one of the most powerful tools for non-stationary signal processing. It is the generalization of many familiar transforms, such as FT, fractional Fourier transform (FrFT) [11,12], Fresnel transform (FrT) [13], and so on. The sampling theories for approximating bandlimited signal in the LCT domain has been proposed in [14–16]. As for non-bandlimited signal in the LCT domain, [17] proposed a new sampling formulate to reconstruct original signals. Liu et al. investigate the

[☆] This work was supposed by the National Science Foundation of China (No. 60901058 and No. 61171195), and also supposed partially by Beijing Natural Science Foundation (No. 1102029).

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sampling formulae for certain types of non-bandlimited signals based on two ladder-shape filters designed in the LCT domain and obtain some important results [18]. But all of these sampling formulae proposed in these papers assume that the location of sampling points are known first. In real application, we often encounter the problem of reconstructing signals when their nonuniform sampling locations are unknown [5]. In this paper, we will provide an algorithm for approximating a signal from nonuniform samples at unknown locations. These signals are not necessarily bandlimited in the FT domain, however bandlimited in the LCT domain. The algorithm proposed in this paper can be looked as the generalization of the classical results in the FT domain.

The paper is organized as follows, in Section 2, we first introduce the LCT of a signal and describe the problem to be solved. A theorem about the linear canonical transform series (LCTS) expansion for finite signals is given in Section 3. Section 4 contains the use of the LCTS expansion as the mathematical model for approximating a signal and a method to estimate the parameters of the model. The simulation results are shown in Section 5. Section 6 contains the conclusion.

2. Preliminary

2.1. The linear canonical transform (LCT)

The LCT with parameter $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of a signal $f(t)$ is defined as follows [8,10]:

$$L_A[f](u) = \begin{cases} \int_{-\infty}^{\infty} f(t) \sqrt{\frac{1}{j2\pi b}} e^{j(j/2)[(a/b)t^2 - (2/b)ut + (d/b)u^2]} dt, & b \neq 0 \\ \sqrt{d} e^{j(j/2)cu^2} f(du), & b = 0 \end{cases} \quad (1)$$

where a, b, c, d are real numbers satisfying $ad - bc = 1$, i.e., $\det(\mathbf{A}) = 1$.

The inverse transform of the LCT (ILCT) is given by the LCT having parameter $\mathbf{A}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Hence, we can obtain the original signal $f(t)$ from $L_A[f](u)$ via

$$f(t) = L_{\mathbf{A}^{-1}}[L_A[f](u)](t) = \begin{cases} \int_{-\infty}^{\infty} L_A[f](u) \sqrt{\frac{j}{2\pi b}} e^{j(j/2)[-(d/b)u^2 + (2/b)ut - (a/b)t^2]} du, & b \neq 0 \\ \sqrt{a} e^{-j(j/2)cat^2} L_A[f](at), & b = 0 \end{cases} \quad (2)$$

From Eq. (1) we can see that, when $b=0$, the LCT of a signal is essentially a chirp multiplication and it is of no particular interest to our object. Therefore, without loss of generality, we set $b > 0$ in the following sections of the paper.

If we chose the parameter $\mathbf{A} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, the LCT will be reduced to the fractional Fourier transform (FRFT) and to Fourier transform (FT) when $\theta = \pi/2$.

A signal $f(t)$ is said to be Ω_A -bandlimited in the LCT sense with parameter $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ [18], if

$$L_A[f](u) = 0 \quad \text{for } |u| > \Omega_A.$$

where Ω_A is the band width of the signal $f(t)$ in the LCT domain.

In [7], the authors indicate that there are lots of signals, which are non-bandlimited in the traditional Fourier domain, however bandlimited in the LCT domain. Under such circumstance, the classical sampling theory associated with FT (FRFT) domain may not work efficiently. Compared with FT and FRFT, the LCT is a more powerful tool in signal processing. Therefore, it is natural to find solution in the LCT domain when problem cannot be solved in the FT (FRFT) domain.

2.2. Outline of the problem

A nonuniform sampling set is a collection of real numbers $\{t_0, t_1, \dots, t_{N-2}, t_{N-1}\}$, with nonuniform increments between the samples. It is a monotonically increasing set of values with the first and last values taken to be $-T/2$ and $T/2$, where T is a real positive value, i.e., $-T/2 = t_0 < t_1 < \dots < t_{N-1} = T/2$. We shall denote the real value of a signal $f(t)$ at these sample locations, $\{f(t_0), f(t_1), \dots, f(t_{N-1})\}$, as $\{f_0, f_1, \dots, f_{N-1}\}$. It is assumed that the first and last sample locations are known, with the remaining sample locations unknown.

Under such circumstance, if a signal is bandlimited in the FT domain, the authors in [5,6] proposed a useful method to estimate the sample locations and reconstruct the original signal separately. But in practical applications, most of the signals are non-stationary and non-bandlimited associated with the FT domain. Therefore it is necessary to propose an optimal method to reconstruct these signals in different transform domain. In this paper, we will probe this problem in the LCT domain in detail.

3. LCT series expansion for finite signals

In order to solve the above problem, we first generalize the classical Fourier series (FS) to the LCT domain. In the traditional FS case, a basis function – sinusoids – has an impulse output in the Fourier domain. In [17], the authors used the same notation to find the fractional Fourier basis functions and developed fractional Fourier series (FRFS). We develop linear canonical transform series (LCTS) in a similar way in the following.

Lemma 1 (Stern [15]). The LCT $L_A[f](u)$ of a compact function such that $f(t) = 0$ for $|x| > T/2$ can be exactly reconstructed from its sampled version $L_A[f_s](u)$ at points $u_n = n(2\pi b/T)$, $n \in Z$ as follows:

$$L_A[f](u) = T_s \cdot L_A \left\{ \text{rect} \left(\frac{x}{T} \right) L_{\mathbf{A}^{-1}} \{ L_A[f_s](u) \}(t) \right\} (u_n)$$

where T_s denotes the sampling interval and

$$L_A[f_s](u) = L_A[f](u) \sum_{n=-\infty}^{\infty} \delta(u - nT_s) = \frac{L_A[f](u)}{T_s} \sum_{k=-\infty}^{\infty} e^{j(2k\pi/T_s)u}$$

Theorem 1. For a finite signal $f(t)$, its linear canonical transform series (LCTS) expansion with parameter $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ can be derived as

$$f(t) = \sum_{n=-\infty}^{\infty} C_{A,n} \sqrt{\frac{j}{T}} e^{j(j/2)[-(d/b)(n2\pi b/T)^2 + (2/b)t(n2\pi b/T) - (a/b)t^2]}$$

and the LCTS expansion coefficients $C_{A,n}$ is defined as

$$C_{A,n} = \int_{-T/2}^{T/2} f(t) \sqrt{\frac{j}{T}} e^{-(j/2)[-(d/b)(n2\pi b/T)^2 + (2/b)t(n2\pi b/T) - (a/b)t^2]} dt.$$

where $t \in [-T/2, T/2]$ and T is the length of the signal.

Proof. The original time domain representation of the impulse function $\delta(u - nu_0)$ in the LCT domain with parameter $\mathbf{A}$ is

$$\begin{aligned} L_{A^{-1}}[\delta(u - nu_0)] \\ &= \int_{-\infty}^{\infty} \delta(u - nu_0) \sqrt{\frac{j}{2\pi b}} e^{(j/2)[-(d/b)u^2 + (2/b)tu - (a/b)t^2]} du \\ &= \sqrt{\frac{j}{2\pi b}} e^{(j/2)[-(d/b)(nu_0)^2 + (2/b)t(nu_0) - (a/b)t^2]} \end{aligned} \quad (3)$$

We define the orthogonal system of LCTS as follows:

$$\Phi_{A,n}(t) = \sqrt{\frac{j}{2\pi b}} e^{(j/2)[-(d/b)(nu_0)^2 + (2/b)t(nu_0) - (a/b)t^2]} \quad (4)$$

The orthogonal system of LCTS is a chirp signal, which is an aperiodic function. We cannot get a periodic signal by such signal, therefore the LCTS is only for finite signals. As for the periodic signal, we can only process it in its one-period.

Compute the integration for any two signals of the orthogonal system along the whole interval $[-T/2, T/2]$,

$$\int_{-T/2}^{T/2} \Phi_{A,m} \Phi_{A,n}^* dt = \begin{cases} \frac{T}{2\pi b}, & m = n \\ \frac{1}{j2\pi(m-n)u_0} e^{(jd/2b)[(nu_0)^2 - (mu_0)^2]} e^{-(j/2b)(m-n)Tu_0} [e^{(j/b)(m-n)Tu_0} - 1], & m \neq n \end{cases} \quad (5)$$

If $\{\Phi_{A,n}\}$ satisfy the orthogonal property, the right of Eq. (5) must be equal to zero when $m \neq n$. This goal can be reached when $u_0 = k2\pi b/T$, where $k \in \mathbb{Z}$. Hence, through dividing each $\Phi_{A,n}$ by $\sqrt{T/2\pi b}$, we can obtain the orthogonal function $\tilde{\Phi}_{A,n}$ of LCTS,

$$\tilde{\Phi}_{A,n}(t) = \frac{\Phi_{A,n}(t)}{\sqrt{\frac{T}{2\pi b}}} = \sqrt{\frac{j}{T}} e^{(j/2)[-(d/b)(nu_0)^2 + (2/b)t(nu_0) - (a/b)t^2]} \quad (6)$$

where $n = -\infty, \dots, -1, 0, 1, \dots, \infty$. Thus $\{\tilde{\Phi}_{A,n}(t)\}$ with $u_0 = k2\pi b/T$ can compose an orthogonal system of LCTS.

Next, we prove the orthogonal system of LCTS $\{\tilde{\Phi}_{A,n}(t)\}$ ($n = -\infty, \dots, -1, 0, 1, \dots, \infty$) is complete when $k=1$.

If a finite signal $f(t)$ is orthogonal with every $\{\tilde{\Phi}_{A,n}(t)\}$, i.e., $\int_{-T/2}^{T/2} f(t) \tilde{\Phi}_{A,n}^* dt = 0$. Then based on the definition of LCT,

$$\begin{aligned} \int_{-T/2}^{T/2} f(t) \tilde{\Phi}_{A,n}^* dt &= \sqrt{\frac{2\pi b}{T}} \int_{-\infty}^{\infty} f(t) \\ &\sqrt{\frac{1}{j2\pi b}} e^{-(j/2)[-(d/b)(nu_0)^2 + (2/b)t(nu_0) - (a/b)t^2]} dt \\ &= \sqrt{\frac{2\pi b}{T}} L_A[f](nu_0) \end{aligned}$$

$$= \sqrt{\frac{2\pi b}{T}} L_A[f] \left(n \frac{2k\pi b}{T} \right)$$

So, $L_A[f](n2k\pi b/T) = 0$. Based on Lemma 1, when $k=1$, we have

$$L_A[f](u) = T_s \cdot L_A \left\{ \text{rect} \left(\frac{x}{T} \right) L_{A^{-1}} \{ L_A[f_s](u) \} (t) \right\} (u_n) = 0$$

Based on ILCT, we see that

$$\begin{aligned} f(t) &= L_{A^{-1}}[L_A[f](u)](t) \\ &= \int_{-\infty}^{\infty} L_A[f](u) \sqrt{\frac{j}{2\pi b}} e^{(j/2)[-(d/b)u^2 + (2/b)ut - (a/b)t^2]} du = 0 \end{aligned}$$

So, the orthogonal system of LCTS $\{\tilde{\Phi}_{A,n}(t)\}$ ($n = -\infty, \dots, -1, 0, 1, \dots, \infty$) is complete when $u_0 = 2\pi b/T$, i.e., $k=1$. Furthermore, when $k=1$, the orthogonal system of LCTS $\{\tilde{\Phi}_{A,n}(t)\}$ ($n = -\infty, \dots, -1, 0, 1, \dots, \infty$) is a basis of LCTS.

The instantaneous frequency of a basis signal is defined as the derivative of its phase, and the phase of $\{\tilde{\Phi}_{A,n}(t)\}$ can be derived as

$$\omega_{A,n}(t) = -(a/b)t + n \frac{2\pi}{T} \quad (7)$$

From Eq. (7), we should see that every basis signal is a chirp signal with chirp rate $-a/b$. Therefore, a finite signal

can be decomposed in terms of its chirp harmonics, i.e., a finite signal $f(t)$ can be reconstructed via Eq. (8),

$$\begin{aligned} f(t) &= \sum_{n=-\infty}^{\infty} C_{A,n} \tilde{\Phi}_{A,n}(t) \\ &= \sum_{n=-\infty}^{\infty} C_{A,n} \sqrt{\frac{j}{T}} e^{(j/2)[-(d/b)(n2\pi b/T)^2 + (2/b)t(n2\pi b/T) - (a/b)t^2]}, \\ t &\in \left[-\frac{T}{2}, \frac{T}{2} \right] \end{aligned} \quad (8)$$

The LCTS expansion coefficients $C_{A,n}$ can be computed by the inner product of the original signal and chirp basis signal,

$$\begin{aligned} C_{A,n} &= \int_{-T/2}^{T/2} f(t) \tilde{\Phi}_{A,n}^* dt \\ &= \int_{-T/2}^{T/2} f(t) \sqrt{\frac{j}{T}} e^{-(j/2)[-(d/b)(n2\pi b/T)^2 + (2/b)t(n2\pi b/T) - (a/b)t^2]} dt \end{aligned} \quad (9)$$

Through Eqs. (8) and (9), an aperiodic signal $f(t)$ on a finite interval, $[-T/2, T/2]$, can be represented as LCTS expansion. The theorem is proved. $\square$

Corollary. The LCTS expansion coefficients $C_{A,n}$ can be obtained from the sampled values of LCT of $f(t)$.

Proof. From Eq. (1), we see that, when $b \neq 0$, the LCT of the signal is defined as

$$L_A[f](u) = \int_{-\infty}^{\infty} f(t) \sqrt{\frac{1}{j2\pi b}} e^{j(2/a)(a/b)t^2 - (2/b)ut + (d/b)u^2} dt \quad (10)$$

Based on Theorem 1, the LCTS coefficients of $f(t)$ can be computed as

$$\begin{aligned} C_{A,n} &= \int_{-T/2}^{T/2} f(t) \tilde{\Phi}_{A,n}^* dt \\ &= \int_{-\infty}^{\infty} f(t) \tilde{\Phi}_{A,n}^* dt \\ &= \int_{-\infty}^{\infty} f(t) \sqrt{\frac{j}{T}} e^{-j(2/a)[-(d/b)(n2\pi b/T)^2 + (2/b)t(n2\pi b/T) - (a/b)t^2]} dt \end{aligned} \quad (11)$$

Comparing Eq. (10) with Eq. (11), we can conclude that

$$\begin{aligned} C_{A,n} &= \int_{-T/2}^{T/2} f(t) \tilde{\Phi}_{A,n}^* dt \\ &= \sqrt{\frac{2\pi b}{T}} \int_{-\infty}^{\infty} f(t) \sqrt{\frac{1}{j2\pi b}} e^{-j(2/a)[-(d/b)(n2\pi b/T)^2 + (2/b)t(n2\pi b/T) - (a/b)t^2]} dt \\ &= \sqrt{\frac{2\pi b}{T}} L_A[f] \left(n \frac{2\pi b}{T} \right) \end{aligned} \quad (12)$$

Therefore the corollary is concluded. $\square$

4. Mathematical model and the gradient descent algorithm

4.1. Mathematical model

As described in Section 3, the LCTS of a finite signal $f(t)$ is defined as

$$\begin{aligned} f(t) &= \sum_{n=-\infty}^{\infty} C_{A,n} \sqrt{\frac{j}{T}} e^{j(2/a)[-(d/b)(n2\pi b/T)^2 + (2/b)t(n2\pi b/T) - (a/b)t^2]} \\ &= \sum_{n=-\infty}^{\infty} C_{A,n} Q(n) e^{j(2\pi n/T)t} e^{-j(a/2b)t^2}, \quad t \in \left[-\frac{T}{2}, \frac{T}{2}\right] \end{aligned} \quad (13)$$

where $Q(n) = \sqrt{j/T} e^{-j(2\pi^2 bd/T^2)n^2}$.

A finite LCTS with a total of $2L+1$ exponentials will be denoted as $f_L(t)$

$$f_L(t) = \sum_{n=-L}^L C_{A,n} Q(n) e^{j(2\pi n/T)t} e^{-j(a/2b)t^2} \quad (14)$$

with T being the total length of the signal.

In the following sections of this paper, we assume a finite signal $f(t)$ is Ω_A -bandlimited in the LCT domain and its sample values $\{f_i\}$, its length T and the index L are all known. The purpose of this paper is to find the unknown sample locations $\{t_i\}$, the coefficients $\{C_{A,n}\}$,

and approximating the signal that gives rise to the samples $\{f_i\}$ via the finite LCTS,

$$f_i = f_L(t_i) = \sum_{n=-L}^L C_{A,n} Q(n) e^{j(2\pi n/T)t_i} e^{-j(a/2b)t_i^2} \quad (i = 1, 2, \dots, N-2)$$

Expanding in matrix form

$$\begin{pmatrix} e_1 & 0 & \dots & 0 \\ 0 & e_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e_{N-2} \end{pmatrix} \begin{pmatrix} z_1^{-L} & \dots & z_1^0 & \dots & z_1^L \\ \vdots & & \vdots & & \vdots \\ z_{N-2}^{-L} & \dots & z_{N-2}^0 & \dots & z_{N-2}^L \end{pmatrix} = \begin{pmatrix} C_{A,-L}Q(-L) \\ \vdots \\ C_{A,0}Q(0) \\ \vdots \\ C_{A,L}Q(L) \end{pmatrix} = \begin{pmatrix} f_1 \\ \vdots \\ f_{N-2} \end{pmatrix} \quad (15)$$

where $e_i = e^{-j(a/2b)t_i^2}$, $z_i = e^{2\pi j t_i/T}$.

For brevity, we rewrite Eq. (15) as $\mathbf{EMC} = \mathbf{f}$. Since the first and last sample locations are supposedly known, these samples are not included in Eq. (15). As talked about in Section 3, the basis function of LCTS is aperiodic, this property confines the LCTS being only applicable for finite signals, i.e., the length of the signal $\mathbf{f}_L(\mathbf{t})$ must be finite. Although the original signal may be periodic with period τ , it is infinite, we are only trying to model the signal from one of its period, $[-T/2, T/2]$.

4.2. Model solution

This part of the section aims to find the sample locations $\{t_i\}$ and the coefficients $\{C_{A,n}\}$ for Eq. (15), thus we can approximate the original signal that gives rise to these samples in the LCT domain via Eq. (14). It can be performed by minimizing the squared difference between the samples and the model to estimate the model's parameters:

$$\min_{\mathbf{EMC}} \|\mathbf{f} - \mathbf{EMC}\|^2 \quad (16)$$

where $\|\cdot\|$ is the l_2 norm.

First, we consider the problem described as

$$\min_{\mathbf{C}} \|\mathbf{f} - \mathbf{EMC}\|^2 \quad (17)$$

Under such circumstance, the nonuniform sample locations are assumed to be known, thus the matrix $\mathbf{B} = \mathbf{EM}$ is fixed. $\mathbf{B}$ is an $(N-2) \times (2L+1)$ matrix.

Actually, the columns of $\mathbf{B}$ are independent when $N-2 \geq 2L+1$. The proof is given as follows:

$$\mathbf{B} = \mathbf{EM} = \begin{pmatrix} e_1 z_1^{-L} & \dots & e_1 z_1^0 & \dots & e_1 z_1^L \\ e_2 z_2^{-L} & \dots & e_2 z_2^0 & \dots & e_2 z_2^L \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ e_{N-2} z_{N-2}^{-L} & \dots & e_{N-2} z_{N-2}^0 & \dots & e_{N-2} z_{N-2}^L \end{pmatrix}$$

When $N-2 \geq 2L+1$, $\mathbf{B}_0$ has the same rank with $\mathbf{B}$, where

$$\mathbf{B}_0 = \begin{pmatrix} e_1 z_1^{-L} & \dots & e_1 z_1^0 & \dots & e_1 z_1^L \\ e_2 z_2^{-L} & \dots & e_2 z_2^0 & \dots & e_2 z_2^L \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ e_{2L+1} z_{2L+1}^{-L} & \dots & e_{2L+1} z_{2L+1}^0 & \dots & e_{2L+1} z_{2L+1}^L \end{pmatrix}$$

$\mathbf{B}_0$ is a invertible matrix. The reason is

$$\begin{aligned} \det(\mathbf{B}_0) &= \det(\mathbf{B}_0^T) \\ &= \det \begin{pmatrix} e_1 z_1^{-L} & e_2 z_2^{-L} & \dots & e_{2L+1} z_{2L+1}^{-L} \\ e_1 z_1^{-L+1} & e_2 z_2^{-L+1} & \dots & e_{2L+1} z_{2L+1}^{-L+1} \\ \vdots & \vdots & \vdots & \vdots \\ e_1 z_1^L & e_2 z_2^L & \dots & e_{2L+1} z_{2L+1}^L \end{pmatrix} \\ &= e_1 z_1^{-L} \dots e_{2L+1} z_{2L+1}^{-L} \end{aligned}$$

$$\cdot \det \begin{pmatrix} 1 & 1 & \dots & 1 \\ z_1 & z_2 & \dots & z_{2L+1} \\ \vdots & \vdots & \vdots & \vdots \\ z_1^{2L} & z_2^{2L} & \dots & z_{2L+1}^{2L} \end{pmatrix}$$

where $\mathbf{B}_0^T$ is the transpose of $\mathbf{B}_0$.

Because $e_i \neq 0$, $z_i \neq 0$, $t_i \neq t_k$ ($i \neq k$), so $\det(\mathbf{B}_0) = e_1 z_1^{-L} \dots e_{2L+1} z_{2L+1}^{-L} \cdot \prod_{1 \leq k < i \leq 2L+1} (z_i - z_k) \neq 0$. This means the columns of $\mathbf{B}_0$ being independent. Furthermore, it means that the columns of $\mathbf{B}$ being independent. So, there is a unique pseudo-inverse matrix for $\mathbf{B}$ when $(N-2) \geq (2L+1)$ (see [19]).

Therefore, further assume N is greater than $(2L+3)$, the unique solution for Eq. (17) exists which is $\mathbf{C} = \mathbf{B}^+ \mathbf{f}$. In the solution, $\mathbf{B}^+ = (\mathbf{B}^H \mathbf{B})^{-1} \mathbf{B}^H$ is the pseudo-inverse of $\mathbf{B}$ and $\mathbf{B}^H$ is the complex conjugate transpose of $\mathbf{B}$. Now, making the substitution $\mathbf{C} = \mathbf{B}^+ \mathbf{f}$ in Eq. (16),

$$\min_{\mathbf{B}} \|\mathbf{f} - \mathbf{B}\mathbf{C}\|^2 \quad (18)$$

After the last iteration, $\mathbf{C} = \mathbf{B}^+ \mathbf{f}$ is held constant. Thus, only $\mathbf{B}$ is variable in Eq. (18).

In addition, if the coefficient vector $\mathbf{C}$ is constant, it is easy to compute the gradient of the cost function with respect to every sample location t_i . We define the i th residual as $R_i = f_i - \mathbf{B}_i \mathbf{C}$, where $\mathbf{B}_i$ is the i th row of $\mathbf{B}$. The cost function, which is defined as the sum of the squares of the residuals, can be expressed as follows:

$$W = \|\mathbf{f} - \mathbf{B}\mathbf{C}\|^2 = \sum_{i=1}^{N-2} R_i^2$$

When the coefficient $\mathbf{C}$ is a constant vector, we compute the derivative of the cost function W ,

$$dW/dt_i = 2R_i dR_i/dt_i = -2R_i (\mathbf{B}'_i \mathbf{C})$$

where $\mathbf{B}'_i$ is the derivative of $\mathbf{B}_i$ with respect to t_i . Then, the next step is updating the sample locations. This step is described in Fig. 1 in detail.

Apply the above step to perform the minimization equation (18), we can get a new iterative time, i.e., a new matrix $\mathbf{B}$ is determined. Thus the new coefficient vector $\mathbf{C}$ can be gained via the equation $\mathbf{C} = \mathbf{B}^+ \mathbf{f}$. Up to now, one circulation of the iterative steps is over.

Based on these knowledge, we can propose a gradient descent algorithm similar to the alternating least-squares algorithm proposed in [6]. It is an alternate iterative algorithm and its flow chart is shown in Fig. 1. By applying the gradient descent algorithm, this paper can acquire the parameters for Eq. (15) and approximate the original signal.

5. Simulation results

A real rectangular function is used in the experiment, it can be seen as a bandlimited signal in the LCT domain. With parameter $\mathbf{A} = \begin{pmatrix} 0.6 & 1.5 \\ 0 & 5 \end{pmatrix}$, where the LCTed signal being a real rectangular function confined to the interval $(-1, 1)$. The LCT pair of the signal is shown in Fig. 2.

As discussed in [14], the signal in Fig. 2 is non-bandlimited in the Fourier domain, therefore the Fourier domain based methods [5,6] cannot be applied here.

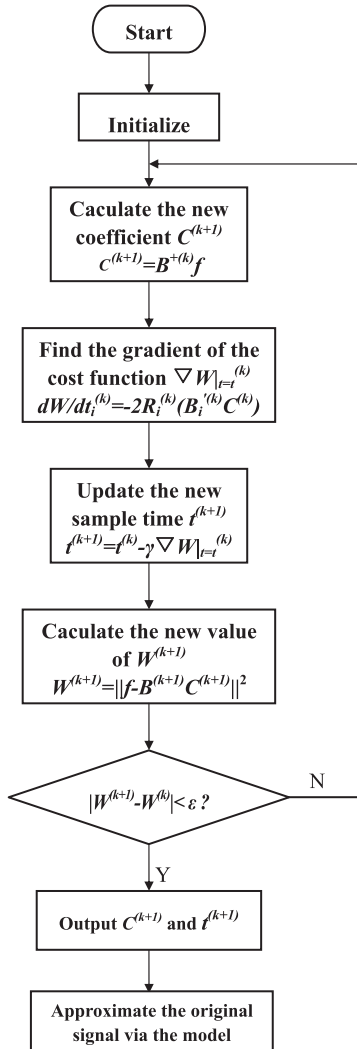


Fig. 1. The flow chart of the gradient descent algorithm (γ is a step size).

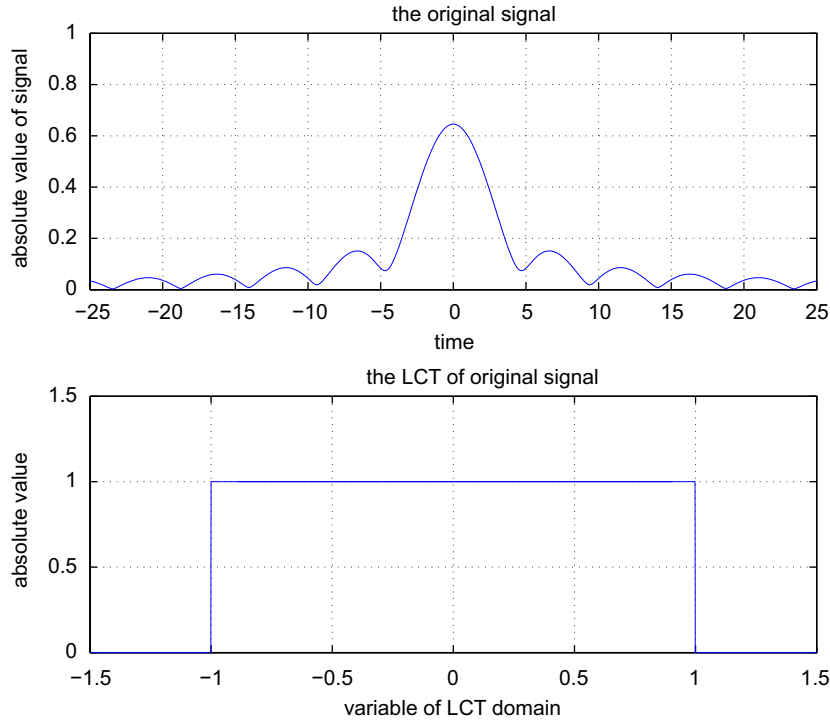


Fig. 2. The LCT pair of the experimental signal.

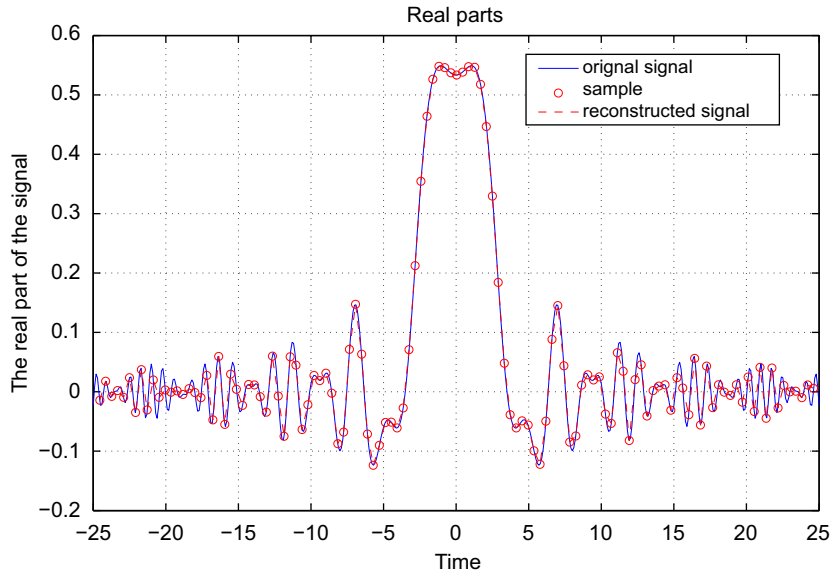


Fig. 3. Real parts of the original, sampling points and the reconstructed signals.

Under such circumstance, we can use the method proposed in this paper to solve the problem.

According to Section 4.1, we apply the finite LCTS as the experimental signal model. The coefficients $\{C_{A,n}\}$ are unknown and the samples gained through adding a random variate on every uniform sample location, i.e., $t_i = t(i) + \varepsilon(i)$ with $t(i)$ being the i th uniform sample location ($i = 1, \dots, N$

–2) and $\varepsilon(i)$ a random variate. Under such circumstance, the sample locations $\{t_i\}$ are unknown. By applying the gradient descent algorithm proposed in Section 4.2, the original signal can be approximated. The main parameters are chosen as follows: $T=50$, $N=123$, $L=60$, $\gamma=0.01$. Figs. 3 and 4 plot the real and imaginary parts of the original, sampling points and the reconstructed signals, respectively.

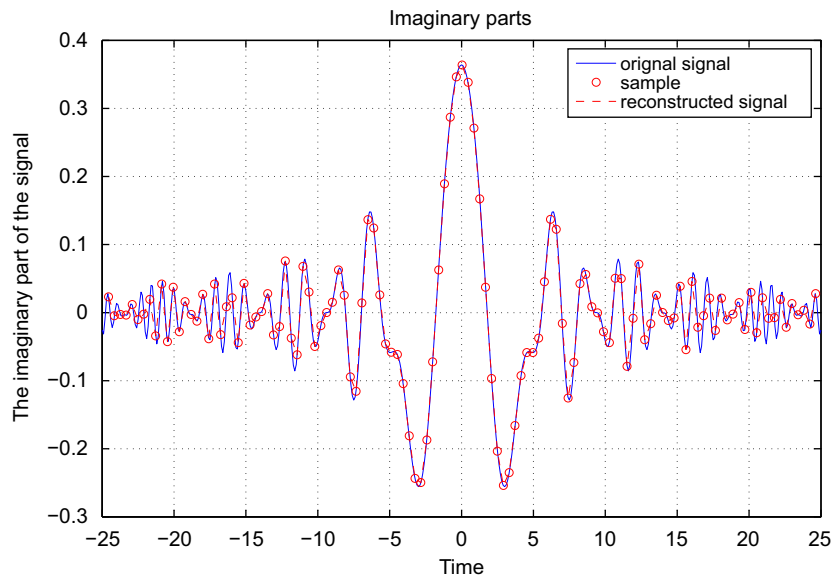


Fig. 4. Imaginary parts of the original, sampling points and the reconstructed signals.

6. Conclusion

Focusing on the problem of approximating signals from nonuniform samples at unknown locations, this paper try investigating it in the LCT domain. We derive the LCTS expansion for the finite signal, apply it as the mathematical model for approximating a signal, provide a gradient descent algorithm to estimate the model's parameters, and finally approximate the original signal. The simulation results are also presented to show the performance of the proposed method.

Based on the properties of LCT and the relationship between LCT and FT, the method proposed in this paper can be used more widely than that proposed in the FT domain.

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